Bogleheads

Risk and return: application

As shown in Risk and return: an introduction, **risk** is a complex topic. But when it comes to *investing,* risk can be summarized as follows:

> *Risk and return are inseparable. Higher returns can only be achieved by taking more risk. There is no free lunch.*

Exceptions are few and far between. The example below uses historical data to demonstrate that, in general, owning stocks is riskier (has more variation in returns) than bonds. You can mitigate (reduce) this risk by diversifying your portfolio.

Variation of returns

As Figure 1 indicates, the return on bonds is less certain than the return on cash (Treasury bills), and the return on stocks is less certain than the return on bonds. Thus, bonds are considered riskier than cash, and stocks are considered riskier than bonds. However, as the risk increases, the average return also increases.

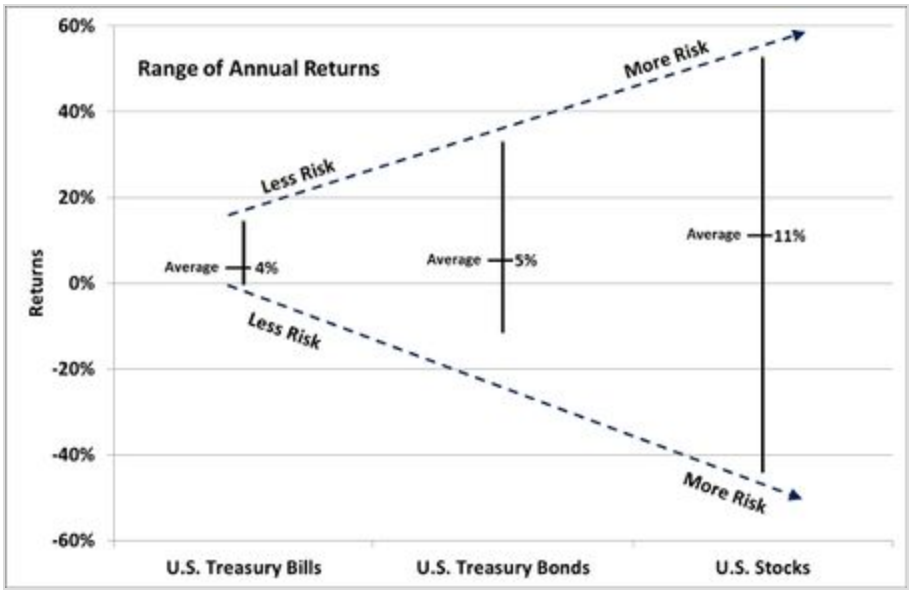


Figure 1. Risk vs. Return: 1928 - 2011[1]

Now, refer to Figure 2, which is Figure 1 but from a different perspective.

As stated in Risk and return: an introduction, *uncertainty* includes both gain *and* loss. Consideration must be given not only for the expected return (*average*), but also for the *range* of variation.

Figure 2 shows this graphically. When the return is higher than planned, a windfall is experienced. A lower return than planned is a shortfall.[note 1]

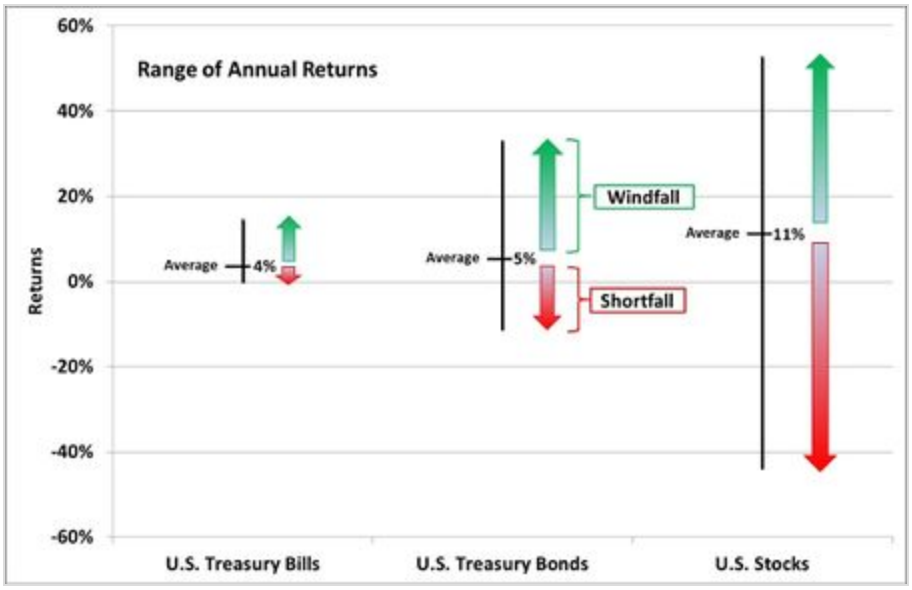


Figure 2. Risk vs. Return for Three Asset Classes: 1928 - 2011[1]

Portfolio diversification

Now, combine those same assets into portfolios. Refer to Figure 3.[note 2]

The Treasury bonds and bills have been combined into a single category called "bonds." This is done intentionally, as asset classes which contain bonds and bills are commonly grouped together as *fixed income*.

Going from left to right, the portfolio progresses from (20% stocks / 80% bonds) to (80% stocks / 20% bonds). Similar to Figure 2, the least variation and lowest return is a portfolio which contains mostly bonds. The highest variation and highest return is a portfolio containing mostly stocks.

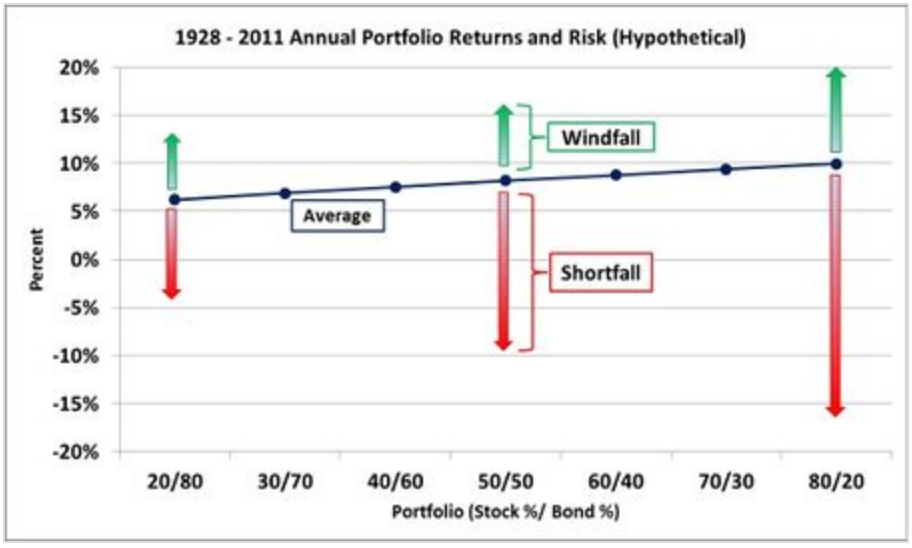


Figure 3. Risk vs. Return for Various Portfolios: 1928 - 2011[1]

This demonstrates that you can reduce your portfolio risk by adding bonds, but this may result in a lower return. Alternatively, you can increase your return by adding stocks, at an increased risk of loss.

Figure 3 shows a *hypothetical* portfolio of assets held from 1928 - 2011 and should not be compared with an actual portfolio. The arrow lengths are used to show relative variation only. Also, do *not* assume that the returns vary evenly (linearly) across the portfolio percentages.[note 3]

Notes

1. Shortfall risk is normally associated with long-term returns and defined in Risk and return: an introduction. Windfall is not a defined risk, but is used to describe an unexpected increase (a "surprise" event).
2. Table 1 shows the portfolio breakdown. The file used to create this table and all the figures in this article can be downloaded from Google Docs: Risk - Historical Performance of Bonds and Stocks.xlsx (https://docs.google.com/open?id=0B8VSNcUR-B6Idk9welFiMDhRX3UybzZxdjdseDVwdw)

Table 1. 1928 - 2011 Portfolio returns and risk (hypothetical)

Portfolio Allocation	**Stocks**	20%	30%	40%	50%	60%	70%	80%
	T.Bills	20%	15%	10%	5%	5%	5%	5%
	T.Bonds	60%	55%	50%	45%	35%	25%	15%
Expected return:		6%	7%	8%	8%	9%	9%	10%

- US stock returns include dividends.
- Three month US Treasury bills
- 10-year US Treasury constant maturity bonds

The Excel file contains additional metrics, including the use of Value at risk to determine the length of the shortfall arrows.

3. It is an effect of this specific hypothetical portfolio held for 83 years; one which cannot be reproduced in the real world.

See also

- Risk and return: an introduction

References

1. "Annual Returns on Stock, T.Bonds and T.Bills: 1928 - Current" (https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html). Damodaran Online.

External links

- The file used to generate the Figures is available as a free download from Google Docs: Risk - Historical Performance of Bonds and Stocks.xlsx (https://docs.google.com/open?id=0B8VSNcUR-B6Idk9welFiMDhRX3UybzZxdjdseDVwdw).

Bibliography

- Benjamin L. Cotton, *The Uncertain Science of Asset Allocation (https://ssrn.com/abstract=182108)*, (September 16, 1999) Available at SSRN.
- Ronald Huisman, Kees C.G. Koedijk, and Rachel A.J. Pownall, *Asset Allocation in a Value-at-Risk Framework (https://ssrn.com/abstract=163970)* (April 27, 1999) Available at SSRN.

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